Twin Layer: Operations & User Guide

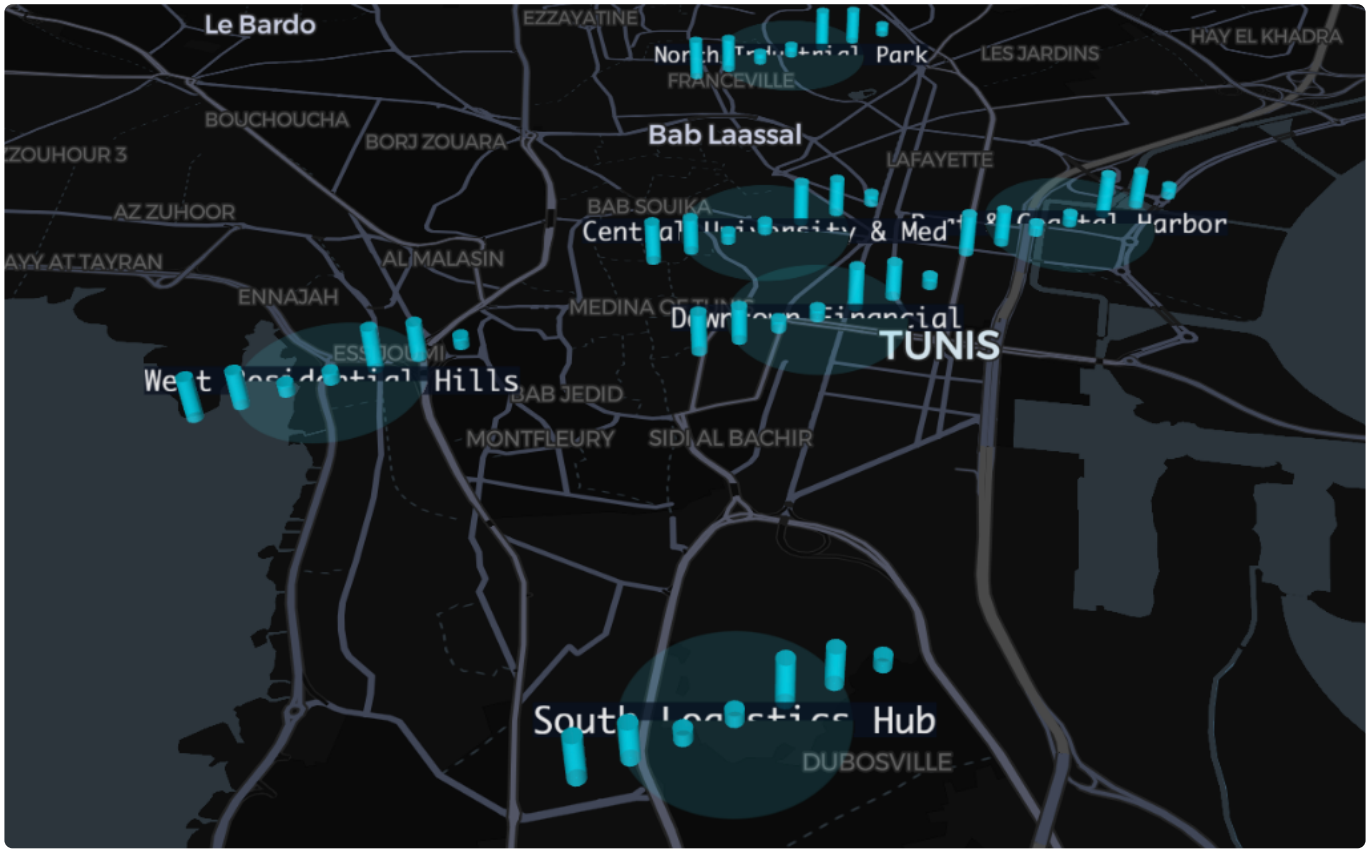
This guide is an illustrated walkthrough of the Twin Layer Adversarial Console. It explains the visual interface, details how each sector and sensor functions, and provides a 5-minute hands-on tutorial for testing all 6 adversarial capabilities.

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1. Interface Tour & Visual System

The Adversarial Console runs in dark tactical mode (`#060913` background with navy glassmorphic panels). The interface uses three status colors:
- **Cyan (`#00e5ff`):** Nominal telemetry and healthy sensor pillars.
- **Amber (`#ffb300`):** Minor variance, unverified key, or degraded connectivity.
- **Crimson (`#ff1744`):** Active adversarial attack, signature failure, or physical state divergence.

A. 3D Tactical Map (Greater Tunis)

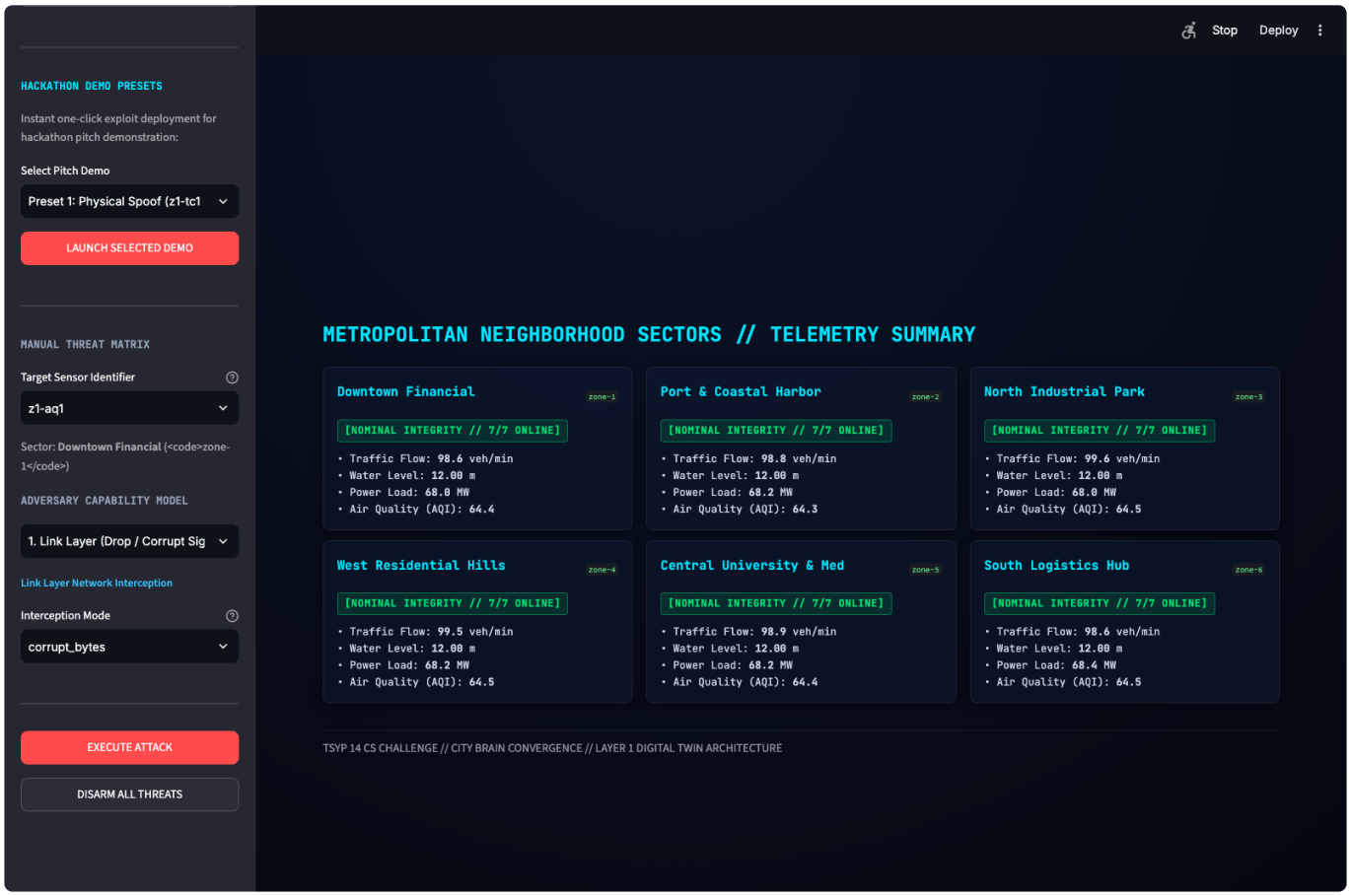


Greater Tunis 3D Tactical Map

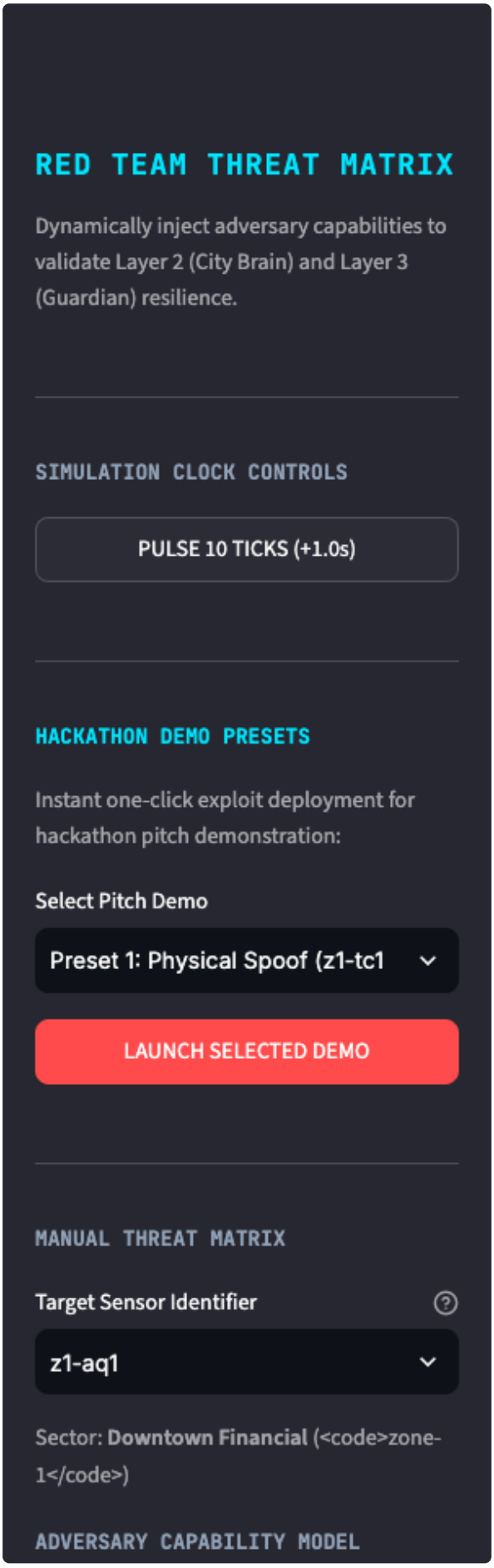
The map displays all 6 municipal sectors across Greater Tunis, with fixed centroids on dry land:
1. **Downtown Financial (`zone-1`):** Commercial center along Avenue Habib Bourguiba.
2. **Port & Coastal Harbor (`zone-2`):** Radès and La Goulette harbor docks and shipping canal.
3. **North Industrial Park (`zone-3`):** Charguia and Ariana Sud industrial hub.
4. **West Residential Hills (`zone-4`):** Residential terrain across Bardo and Manouba hills (positioned north of Sebkhet Sijoumi lake).
5. **Central University & Med (`zone-5`):** El Manar campus and Rabta medical district.
6. **South Logistics Hub (`zone-6`):** Ben Arous freight rail terminal and supply warehouses.

Each sensor is rendered as a 3D hexagonal column whose height scales with its current metric. When a sensor is nominal, its pillar and sector halo glow cyan. When an attack is injected, the target pillar elevates and turns crimson, surrounded by a red warning beacon.

B. Neighborhood Operational Cards & Red Team Sidebar



Metropolitan Neighborhood Sectors and Telemetry Summary



Red Team Threat Matrix Controls

- **Neighborhood Cards (Main View):** Each card summarizes the 7 sensors operating within that zone:
- **Traffic Cameras (`tc1` , `tc2`):** Vehicle passage rates (0 - 200 veh/min).
- **Water Level Floats (`wl1` , `wl2`):** Drainage channel depth (0 - 12 meters).
- **Power Load Meters (`pl1`):** Electrical grid draw (5 - 120 MW).
- **Air Quality Stations (`aq1`):** Particulate index (10 - 350 AQI).
- **Emergency Dispatch (`ec1`):** Active incoming emergency calls (0 - 30 call/min).
- **Red Team Controls (Sidebar):** Contains attack configuration controls, one-click demo presets, simulation clock pulse controls, and the active threat manifest showing all currently running exploits.

C. Clean Vector Cartography



Clean Vector Cartography

The basemap uses clean dark vector tiles (`CARTO Dark Matter`) without cluttered text labels, allowing clean zooming into street-level intersections without broken fonts or overlapping water bodies.

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2. Hands-On Tutorial: Testing the System in 5 Minutes

Follow these steps to test every capability in real time.

Step 1: Open the Console & Verify Nominal Baseline

1. Open your browser and navigate to `http://localhost:8501` .
2. Inspect the top Scoreboard:

3. **Total Readings:** Actively incrementing.
4. **Active Incidents / Tampered:** 0.
5. **Active Red Team Attacks:** 0 (`Zero Threats`).
6. **System Status:** `[SYSTEM STATUS: AUTONOMOUS] // LOCAL SIMULATION RUNTIME ACTIVE` .
7. Verify that all 6 neighborhood cards show green `[NOMINAL INTEGRITY // 7/7 ONLINE]` badges.

Step 2: Inject a Physical Sensor Spoof (`device_physical`)

In this attack, the physical sensor reading is falsified on the wire, but the ground-truth database records the true physical reality.

1. In the sidebar under **"MANUAL THREAT MATRIX"**:
2. Set **Target Sensor Identifier** to `z1-tc1` (Downtown Traffic Camera 1).
3. Set **Adversary Capability Model** to `4. Physical Layer (Sensor Spoofing)` .
4. Enter **Spoofed Telemetry Value**: `185.0` (simulating extreme artificial traffic gridlock).
5. Click **"EXECUTE ATTACK"**.

What happens immediately:
- The Downtown Financial card flips to red: `[CRITICAL THREAT ACTIVE]` .
- On the 3D map, the `z1-tc1` pillar elevates and pulses crimson.
- The sidebar **Active Threat Manifest** lists `[THREAT ACTIVE] device_physical (z1-tc1)` .

Step 3: Inspect Ground Truth vs Broadcast Divergence

1. Click **Tab 2: "Live Telemetry & Divergence Curves"**.
2. Select Device `z1-tc1` from the dropdown.
3. Observe the Altair time-series chart:
4. **Solid Emerald Line:** Real physical reality (e.g., ~45 veh/min).
5. **Dotted Crimson Line:** Spoofed broadcast value (185 veh/min).
6. **Divergence Metric:** Shows an absolute discrepancy of ~140 veh/min.
7. This divergence proves the physical isolation rule: the external world receives the poisoned data, but the Twin's internal SQLite ledger preserves true ground reality.

Step 4: Test the Other 5 Attack Capabilities

You can test each capability individually using the sidebar controls:

Capability	Target Device	Test Configuration	Observable Result
`device_key` (Compromised Key)	`z3-pl1` (Power Load)	Replaces device key with an attacker key	Outbound signature remains valid, but Guardian detects revoked/unauthorized public key on MCP.
`link` (Network Tampering)	`z2-wl1` (Water Float)	Corrupts raw JSON bytes or drops frames	Guardian signature validation immediately fails (`Ed25519 verification failed`); payload corrupted.
`rogue` (Unregistered Transmitter)	`zone-4` (Residential)	Transmits unauthorized packet `z4-rogue-1`	Packet appears on wire; Guardian MCP tool `list_devices` flags sensor as unregistered intruder.

Capability	Target Device	Test Configuration	Observable Result
`sensor_fault` (Hardware Drift)	`z5-aq1` (Air Quality)	Injects additive drift (`+45.0 AQI`)	Values drift upward without triggering hard boundaries; caught by multi-sensor neighborhood correlation.
`actuator` (Command Disobedience)	`z1-wl2` (Flood Valve)	Sets actuation state to `stuck_closed`	Device returns `"status": "ok"` to City Brain MCP call, but physically refuses to open drainage valve.

Step 5: Use One-Click Hackathon Pitch Presets

During live presentations, you do not need to manually configure parameters. The sidebar includes **6 Pitch Presets** pre-calibrated for demonstration:

1. Click **"DEMO PRESET 1: PHYSICAL SPOOF (DOWNTOWN)"**.
2. Spikes traffic camera `z1-tc1` to 195 veh/min.
3. Demonstrates immediate detection on 3D map.
4. Click **"DEMO PRESET 2: FORGED KEY (INDUSTRIAL)"**.
5. Emulates key theft on power substation `z3-pl1`.
6. Demonstrates zero-trust key revocation.
7. Click **"DEMO PRESET 3: LINK CORRUPTION (HARBOR)"**.
8. Injects bitflips into harbor water level `z2-wl1`.
9. Demonstrates signature verification failure.

Step 6: Disarm Threats & Confirm Baseline Recovery

1. In the sidebar, click the button **"DISARM ALL THREATS"**.
2. **Verification:**
3. Active attacks count immediately resets to `0` (`Zero Threats`).
4. All 6 neighborhood cards turn back to `[NOMINAL INTEGRITY // 7/7 ONLINE]`.
5. All 3D pillars return to uniform cyan baseline.
6. The Active Threat Manifest clears completely.

Step 7: Forensic Audit Ledger

1. Click **Tab 3: "Raw Ledger & Forensic Audit"**.
2. The table shows the immutable record of all injected attacks stored in `attacks_injected`:
3. Exact simulation timestamp (`start_sim_time` and `end_sim_time`).
4. Attack type, target device, and attack parameters.
5. Active status (`1` when running, `0` once disarmed).
6. This audit trail gives judges and security evaluators proof of attack lifecycles and response timestamps.